

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	57800	52.208473
1	[Git, Change, Create]	16910	15.274140
2	[Git, Commit]	11176	10.094842
3	[Git, Change, Delete]	8703	7.861078
4	[File, Git]	5776	5.217234
5	[Git, Change, Rename]	3315	2.994309
6	[Git, Tag]	1703	1.538253
7	[Author, Git, Person]	1261	1.139012
8	[Json, Key]	668	0.603378
9	[Json, Value, Scalar]	603	0.544666
10	[Committer, Git, Person]	370	0.334206
11	[NPM, Dependency]	338	0.305302
12	[Type, TS, Primitive]	291	0.262849
13	[Type, TS, Declared]	277	0.250203
14	[TS, ExternalDeclaration]	215	0.194201
15	[Type, TS, Literal]	136	0.122843
16	[Json, Value, Object]	133	0.120134
17	[Type, TS, Union]	119	0.107488
18	[Type, TS, ObjectMember]	102	0.092133
19	[Git, Change, Copy]	102	0.092133
20	[NPM, Script]	91	0.082197
21	[TS, Property]	65	0.058712
22	[TS, Function]	47	0.042453
23	[Type, TS, FunctionParameter]	40	0.036130
24	[Git, Branch]	40	0.036130
25	[Type, Object, TS]	39	0.035227
26	[File, Directory]	34	0.030711
27	[Type, TS, Function]	34	0.030711
28	[TS, Parameter]	33	0.029808
29	[Package, File, Json, NPM]	29	0.026195

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

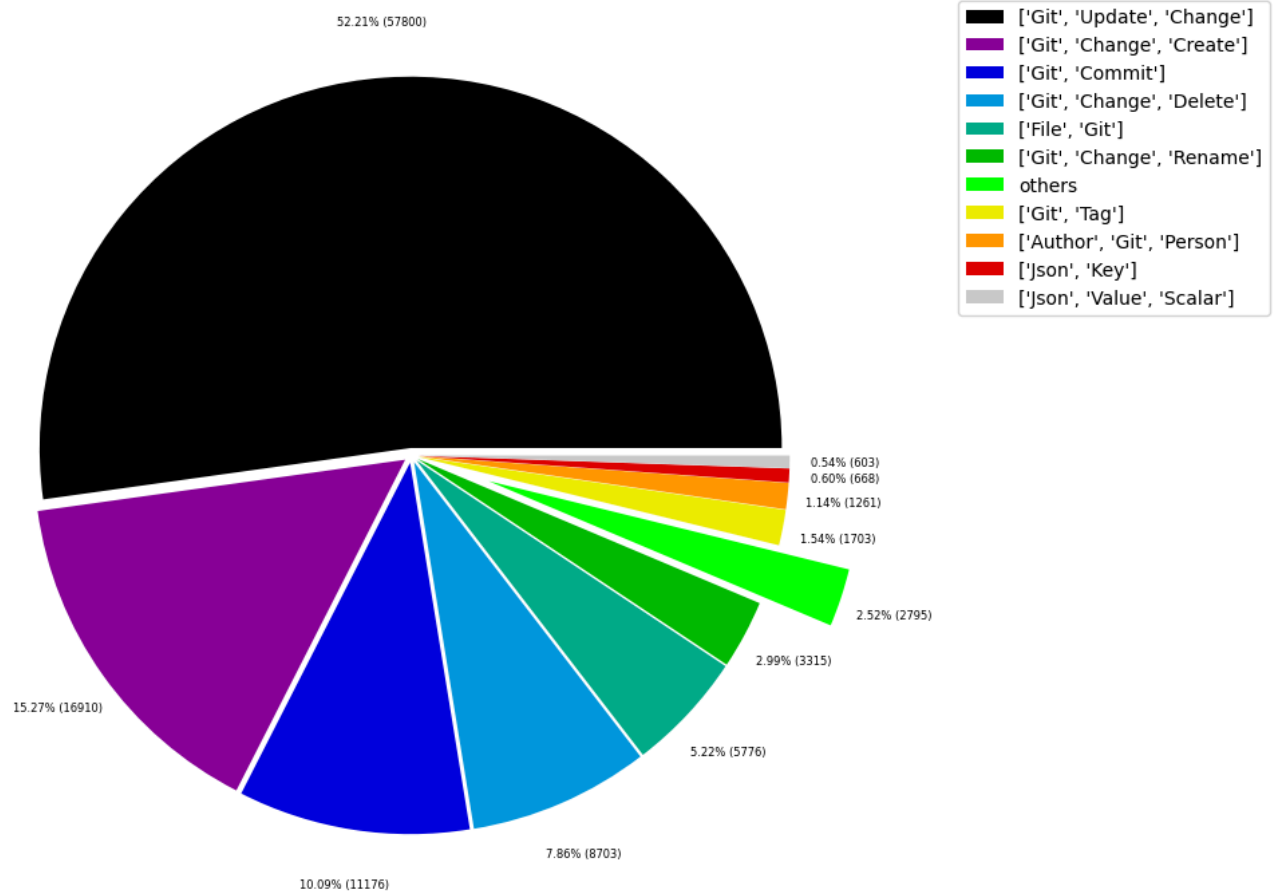


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000903
1	[NPM, PublishConfig]	1	0.000903
2	[File, TS, Scan]	1	0.000903
3	[TS, Method]	1	0.000903
4	[Repository, File, Git]	1	0.000903
5	[TS, Constructor]	1	0.000903
6	[Value, TS, ObjectMember]	1	0.000903
7	[TS, Class]	1	0.000903
8	[TS, Enum]	2	0.001807
9	[Value, Object, TS]	3	0.002710
10	[Type, TS, Tuple]	3	0.002710
11	[Value, TS, Function]	4	0.003613
12	[TS, TypeParameter]	4	0.003613
13	[Value, TS, Complex]	5	0.004516
14	[Repository, NPM]	5	0.004516
15	[NPM, Engine]	6	0.005420
16	[Project, TS]	6	0.005420
17	[File, Local]	6	0.005420
18	[Type, TS, TypeParameterReference]	6	0.005420
19	[Value, TS, Member]	6	0.005420
20	[File, TS, Local, Module]	6	0.005420
21	[Value, TS, Call]	6	0.005420
22	[TS, EnumMember]	8	0.007226
23	[Type, TS, NotIdentified]	11	0.009936
24	[TS, ExternalModule]	11	0.009936
25	[Json, Value, Array]	12	0.010839
26	[Value, TS, Declared]	13	0.011742
27	[TS, TypeAlias]	16	0.014452
28	[File, Directory, Local]	16	0.014452
29	[TS, Interface]	17	0.015355

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

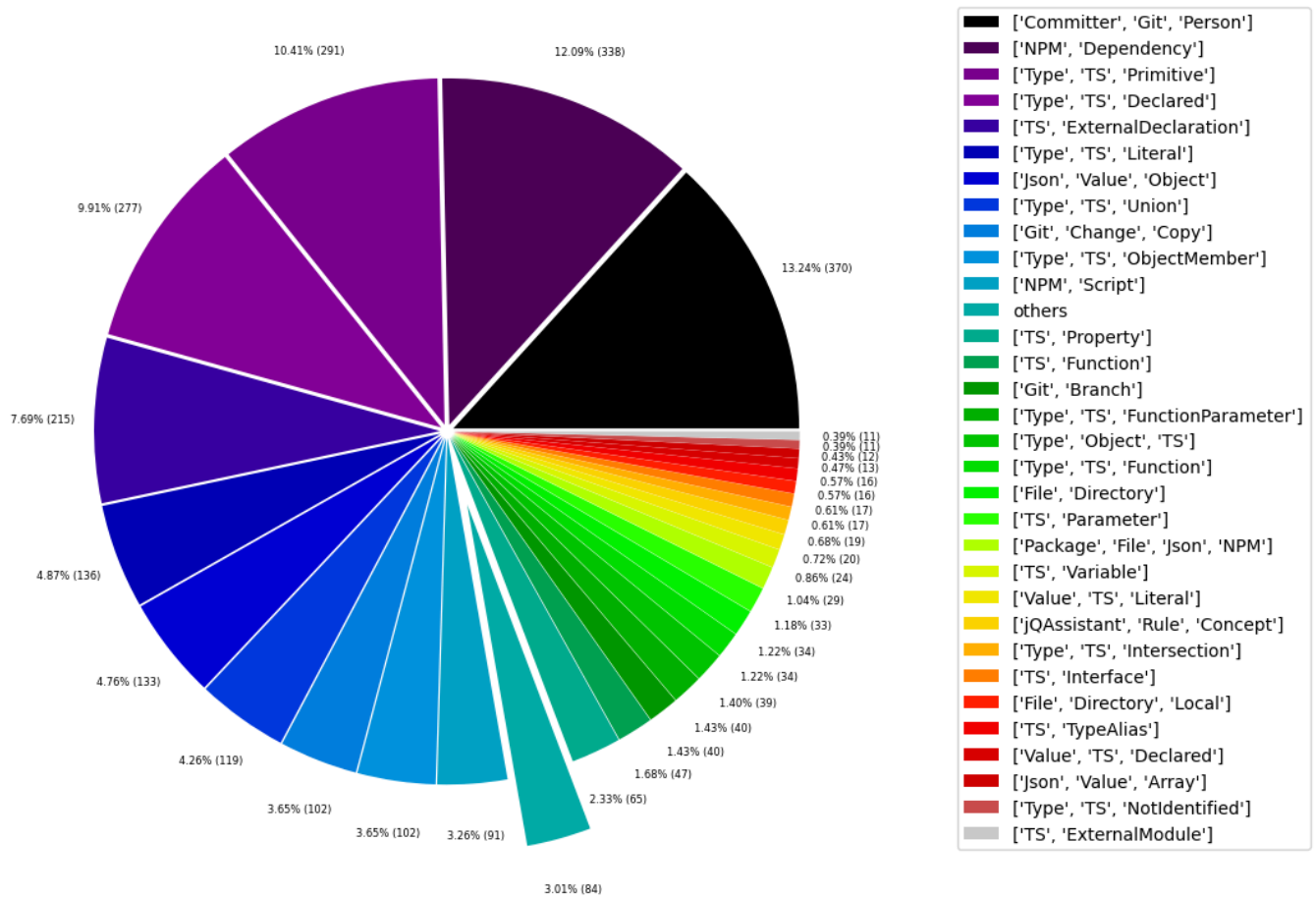


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

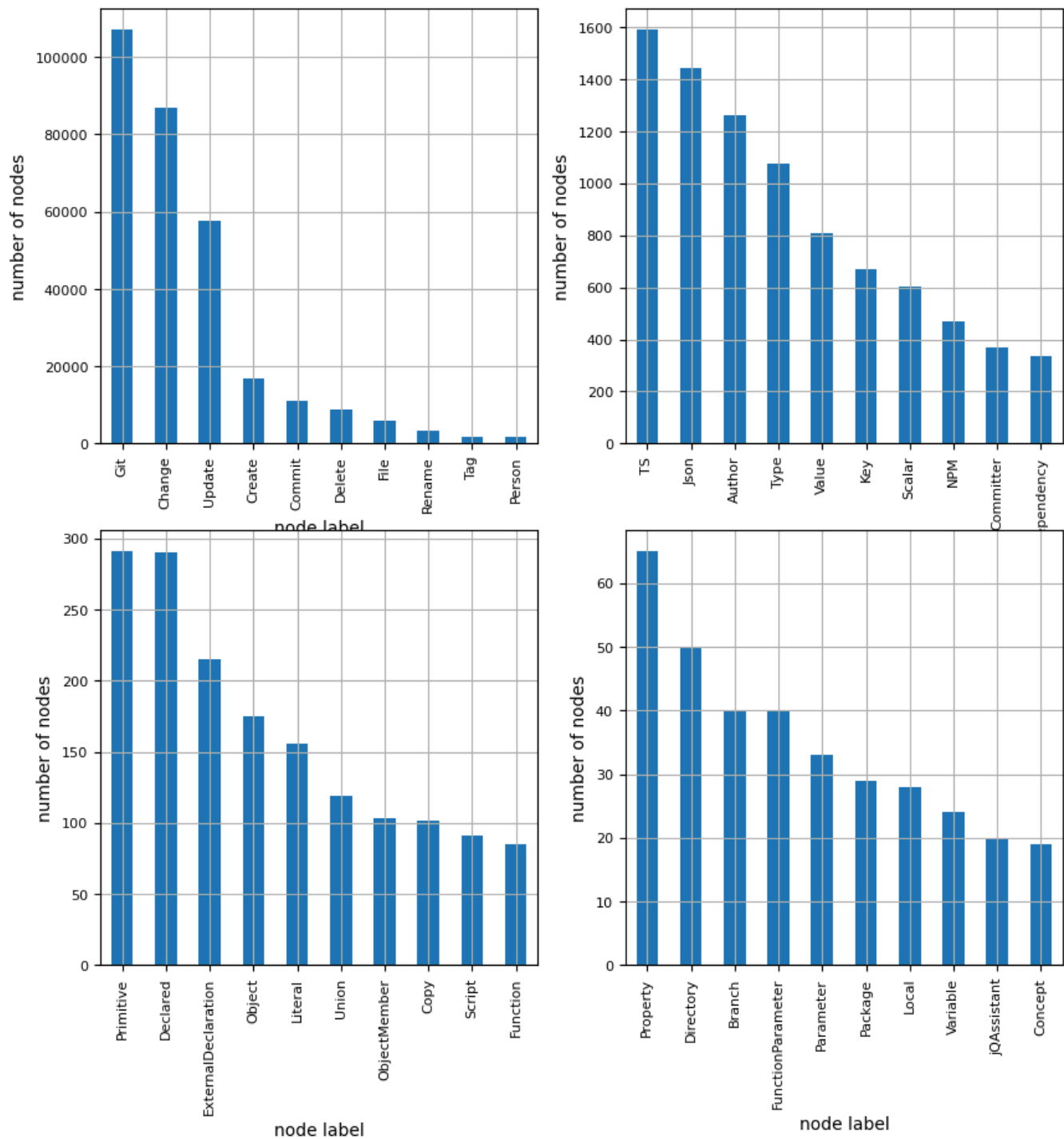
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	107157	96.790714
1	Change	86830	78.430133
2	Update	57800	52.208473
3	Create	16910	15.274140
4	Commit	11176	10.094842
5	Delete	8703	7.861078
6	File	5869	5.301237
7	Rename	3315	2.994309
8	Tag	1703	1.538253
9	Person	1631	1.473218
10	TS	1591	1.437088
11	Json	1445	1.305212
12	Author	1261	1.139012
13	Type	1075	0.971005
14	Value	806	0.728028
15	Key	668	0.603378
16	Scalar	603	0.544666
17	NPM	470	0.424533
18	Committer	370	0.334206
19	Dependency	338	0.305302
20	Primitive	291	0.262849
21	Declared	290	0.261946
22	ExternalDeclaration	215	0.194201
23	Object	175	0.158071
24	Literal	156	0.140909
25	Union	119	0.107488
26	ObjectMember	103	0.093036
27	Copy	102	0.092133
28	Script	91	0.082197
29	Function	85	0.076777
30	Property	65	0.058712
31	Directory	50	0.045163
32	Branch	40	0.036130
33	FunctionParameter	40	0.036130
34	Parameter	33	0.029808
35	Package	29	0.026195
36	Local	28	0.025291
37	Variable	24	0.021678
38	jQAssistant	20	0.018065
39	Concept	19	0.017162

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 330628

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	86830	26.262144
1	MODIFIES	86830	26.262144
2	UPDATES	57800	17.481883
3	COMMITTED	22352	6.760468
4	CREATES	20327	6.147997
5	HAS_PARENT	12251	3.705373
6	DELETES	12018	3.634901
7	HAS_COMMIT	11176	3.380234
8	HAS_FILE	5776	1.746978
9	RENAMES	3315	1.002637
10	HAS_NEW_NAME	1763	0.533228
11	HAS_TAG	1703	0.515080
12	ON_COMMIT	1703	0.515080
13	HAS_AUTHOR	1261	0.381395
14	DEPENDS_ON	887	0.268277
15	HAS_KEY	668	0.202040
16	HAS_VALUE	668	0.202040
17	CONTAINS	594	0.179658
18	HAS_COMMITTER	370	0.111908
19	OF_TYPE	338	0.102230
20	EXPORTS	309	0.093459
21	REFERENCES	197	0.059584
22	DECLARES	186	0.056257
23	DECLARES_DEV_DEPENDENCY	169	0.051115
24	DECLARES_DEPENDENCY	161	0.048695
25	HAS_MEMBER	103	0.031153
26	COPIES	102	0.030850
27	HAS_TYPE_ARGUMENT	94	0.028431
28	DECLARES_SCRIPT	91	0.027523
29	RETURNS	82	0.024801

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

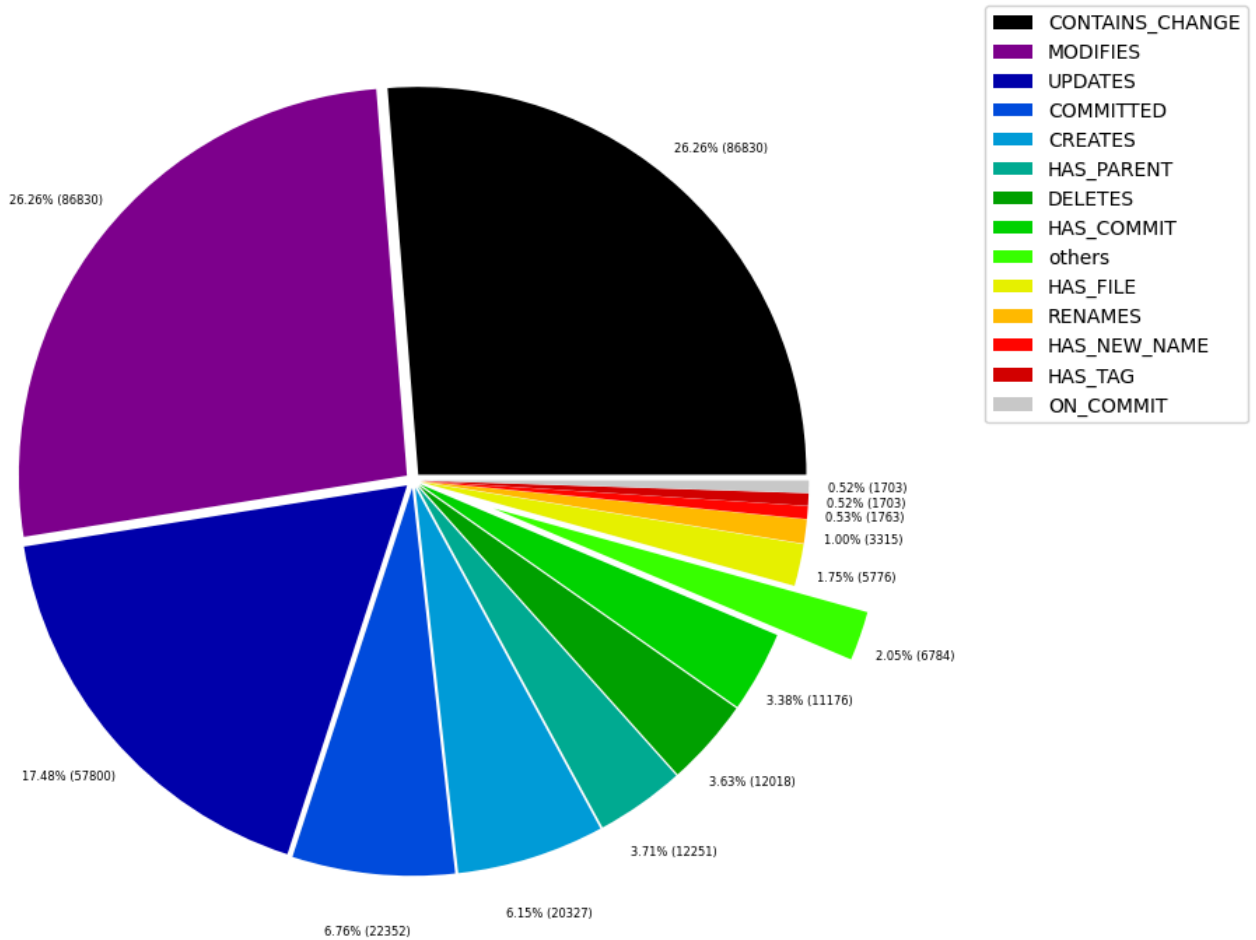


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

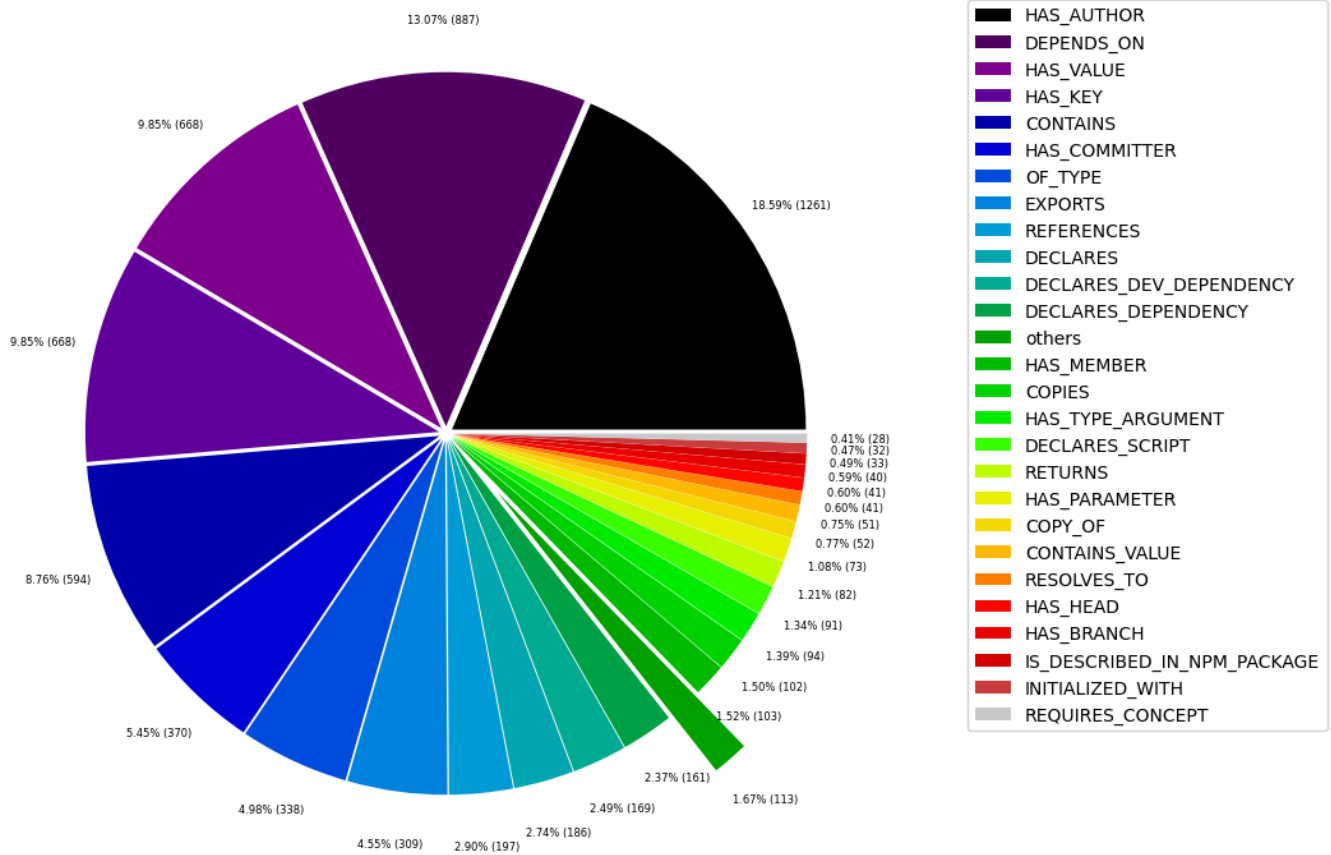
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000302
1	CONSTRAINED_BY	4	0.001210
2	IN_REPOSITORY	5	0.001512
3	REFERENCED_PROJECTS	5	0.001512
4	PARENT	6	0.001815
5	MEMBER	6	0.001815
6	HAS_ROOT	6	0.001815
7	HAS_NPM_PACKAGE	6	0.001815
8	HAS_CONFIG	6	0.001815
9	HAS_ARGUMENT	6	0.001815
10	EXTENDS	6	0.001815
11	DECLARES_ENGINE	6	0.001815
12	CONTAINS_PROJECT	6	0.001815
13	CALLS	6	0.001815
14	DECLARES_PEER_DEPENDENCY	8	0.002420
15	USES	11	0.003327
16	INCLUDES_CONCEPT	19	0.005747
17	REQUIRES_CONCEPT	28	0.008469
18	INITIALIZED_WITH	32	0.009679
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009981
20	HAS_BRANCH	40	0.012098
21	RESOLVES_TO	41	0.012401
22	HAS_HEAD	41	0.012401
23	CONTAINS_VALUE	51	0.015425
24	COPY_OF	52	0.015728
25	HAS_PARAMETER	73	0.022079
26	RETURNS	82	0.024801
27	DECLARES_SCRIPT	91	0.027523
28	HAS_TYPE_ARGUMENT	94	0.028431
29	COPIES	102	0.030850

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	UPDATES	[File, Git]	57800	57800	
1	[Git, Update, Change]	MODIFIES	[File, Git]	57800	57800	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	57800	11176	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	16910	11176	
4	[Git, Change, Create]	MODIFIES	[File, Git]	16910	16910	
5	[Git, Change, Create]	CREATES	[File, Git]	16910	16910	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12251	11176	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11176	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11176	1261	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11176	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8703	11176	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8703	8703	
12	[Git, Change, Delete]	DELETES	[File, Git]	8703	8703	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5776	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3315	11176	
15	[Git, Change, Rename]	CREATES	[File, Git]	3315	3315	
16	[Git, Change, Rename]	MODIFIES	[File, Git]	3315	3315	
17	[Git, Change, Rename]	DELETES	[File, Git]	3315	3315	
18	[Git, Change, Rename]	RENAMES	[File, Git]	3315	3315	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1763	5776	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1703	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1703	1703	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1261	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

```

total_number_of_nodes (vertices): 110710
total_number_of_relationships (edges): 330628
-> total directed graph density: 2.6975521621673052e-05
-> total directed graph density in percent: 0.0026975521621673054

```